

JUNLANG HUANG

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I. EDUCATION

Carnegie Mellon University

M.S. in Artificial Intelligence Engineering – Electrical and Computer Engineering

Pittsburgh, PA, USA

Aug. 2026 – Present

Sun Yat-sen University

B.Eng. in Microelectronic Technology and Engineering

Guangzhou, China

Sep. 2022 – Jun. 2026

- **GPA:** 3.80/4.00; University Scholarship (2023, 2024).

II. RESEARCH INTERESTS

- **Research Interests** — AI for EDA; learning-based circuit analysis and simulation; semiconductor data analytics; structured circuit representations; physical-design optimization. Prospective interests include LLM-based EDA and data-driven IC test and diagnosis.

III. PUBLICATIONS

* *Equal contribution*

- **J. Huang***, H. Chen*, L. Luo, Y. Cai, L. Zhang, T. Ma, Y. Zhang, Z. Guan, “S-Crescendo: A Nested Transformer Weaving Framework for Scalable Nonlinear System in S-Domain Representation,” **NeurIPS 2025**.
Contribution: Co-first author; designed the neural architecture, implemented training and inference, built HSPICE and NGSPICE automation plus data pipelines, and ran circuit-level experiments.
- H. Chen, **J. Huang**, Z. Sha, Z. Wu, S. Zhang, Z. Jin, C. Zhuo, “RA²L-NAND: Risk-Aware Active Learning for High-Sigma Exploration in 3D NAND,” **ICCAD 2026 (accepted)**.
Contribution: Worked on semiconductor-domain active learning for high-sigma / rare-region exploration; built the experiment and evaluation pipeline and analyzed limited-budget sampling behavior.
- **J. Huang***, H. Chen*, Z. Guan, “S-Crescendo: A Nested Transformer Weaving Framework for Scalable Nonlinear System in S-Domain Representation,” **IEEE TCAD, under review**.
Contribution: Extending the conference work with broader experiments, stronger cross-corner validation, and expanded technical analysis.

IV. RESEARCH EXPERIENCE

SYSU EDA Lab, Sun Yat-sen University

Guangzhou, China

Role: Research Assistant; Supervisor: Prof. Zhong Guan

Sep. 2023 – Jun. 2026

- Conducted research in machine learning for EDA and semiconductor systems, with responsibilities spanning problem formulation, data generation, algorithm design, implementation, experimentation, and manuscript preparation.

1. S-Crescendo: Learning-Based Circuit Transient Modeling

Guangzhou, China

Role: Co-first Author, Lead Developer; SYSU EDA Lab

Nov. 2024 – May 2025

- Developed an S-domain neural model for transient simulation of a nonlinear driver plus a linear RC load.
- Implemented an autoregressive Transformer in PyTorch and built automated HSPICE/NGSPICE workflows for data generation, evaluation, and circuit-level numerical experiments.
- Achieved R^2 up to 0.997 against HSPICE and generalized from training orders ≤ 10 to unseen orders up to 200 ($R^2 \approx 0.98\text{--}0.99$), with $6\text{--}19\times$ speedup over HSPICE.

2. RA²L-NAND: Active Learning for High-Sigma Exploration in 3D NAND

Guangzhou / Hangzhou, China

Role: Research Collaborator; Collaboration with Zhejiang University

2025 – 2026

- Worked on risk-aware active learning for high-sigma / rare-region exploration in 3D NAND under limited sampling budgets.

- Built the experiment and evaluation pipeline, including data processing, sample-selection analysis, and characterization of rare regions.

3. Natural-Language-to-Analog-Circuit Netlist Generation

Guangzhou, China

Role: Project Leader (Research / Engineering); SYSU EDA Lab

Aug. 2025 – Present

- Explored natural-language-conditioned generation of analog circuit netlists as a step toward language-guided circuit design automation.
- Used **hypergraph** representations to model multi-terminal devices and shared-node relations more faithfully than ordinary pairwise graphs, with emphasis on circuit topology preservation.
- Built preprocessing and modeling experiments to improve structural validity and to map design intent into structured circuit representations.

4. RL-Guided Placement ECO for Signoff Power Integrity

Guangzhou, China

Role: Undergraduate Thesis Author

Sep. 2025 – Jun. 2026

- Developed an end-to-end reinforcement-learning pipeline for signoff-stage placement ECO under industrial backend constraints.
- Integrated **Innovus**, **PrimeTime**, and **RedHawk** with PPO-based optimization, feature engineering, PyTorch training, and RLlib-based parallel simulation.
- Reduced worst-case **static IR drop** by nearly **30%** on representative SoC blocks while simultaneously reducing timing violations; this project formed my undergraduate honors thesis.

V. LEADERSHIP

AI Research Group Lead, SYSU EDA Lab — Coordinated a small AI-for-EDA research subgroup; assigned tasks, supported collaborative execution, and helped organize manuscript preparation.

Audio Amplifier Design Lead — Led a small hardware project spanning circuit design, PCB layout, enclosure prototyping, and team coordination; shipped more than one hundred units.

VI. ENGINEERING SKILLS

EDA / Circuit — Cadence Virtuoso, HSPICE, NGSPICE, PrimeTime, RedHawk, Innovus, Sentaurus TCAD.

ML / Programming — Python, PyTorch, RLlib, C/C++, MATLAB.